1. Tokenization: { BPE algo.
   MED. / RE

2. Tokenizer { learner
   segmenter

3. Download dataset { hf-mirror.com
   modelscope

① p_{data} / p_θ

$W_{1:n} = (w_1, w_2, \cdots, w_n)$, $w_{1:n} \sim p_{data}(\cdot)$

$D = \{ w^{(i)} \}_{i=1}^{N}$: dataset $\begin{cases} \text{training data} \\ \text{testing data} \end{cases}$

$w^{(i)} \sim p_{data}(\cdot)$

Find $p_\theta \approx p_{data}$: unknown.

Training : make $p_\theta \approx p_{data} : P$

$\theta^* \in \arg\min_\theta D(p_{data}, p_\theta)$ s.t.

$p_{\theta^*}(w) \approx p_{data}(w)$

choices of D :

(1). KL-Divergence (most common)

$$D_{KL}(p_{data} \| p_\theta) = \int p_{data}(w) \cdot \log \frac{p_{data}(w)}{p_\theta(w)} \, dw$$

$$= \mathbb{E}_{w \sim p_{data}} [\log p_{data}(w) - \log p_\theta(w)]$$

- $D_{KL}(p \| q) \neq D_{KL}(q \| p)$
- $D_{KL}(p \| q) \geq 0$
- $D_{KL}(p \| q) = 0$ iff $p = q$.

$p = (0.5, 0.5)$. $q = (0.9, 0.1)$

$\begin{cases} KL(p \| q) = 0.511 \\ KL(q \| p) = 0.367 \end{cases}$

(2) Fisher-Divergence / ...

$$\min_{\theta} D_{KL}(p_{data} \| p_\theta)$$

$$= \min_{\theta} \mathbb{E}[\log p_{data}] + \min_{\theta} \mathbb{E} \log p_\theta(w)$$

$$\arg\min_{\theta} D_{KL}(p_{data} \| p_\theta)$$

$$= \arg\max_{\theta} \mathbb{E}[\log p_\theta(w)].$$

$$\max_{\theta} \frac{1}{N} \sum_{i=1}^{N} \log \underline{p_\theta(w^{(i)})}. \qquad w^{(i)} \sim p_{data}$$

How to calculate it ?

$$p_\theta(w) = \prod_{t=1}^{n} p_\theta(w_t \mid w_{1:t-1})$$

$$p_\theta(x_1, x_2, \cdots, x_N)$$

$$= p_\theta(x_N \mid x_1 x_2 \cdots x_{N-1}) \cdot p(x_1 x_2 \cdots x_{N-1})$$

$$= p_\theta(\cdots) \cdot p_\theta(x_{N-1} \mid x_1 x_2 \cdots x_{N-2})$$

$$\vdots$$

$$= \prod_{t=1}$$

LM: $\begin{cases} \cdot \text{ vocab}: V \\ \cdot \text{ t-length sentence}: w_{1:t},\ w_t = \text{EOS} \end{cases}$

V^+: set of sentences.

A language model is a probability distribution function $p(w_{1:t})$ over V^+ such that

- For any $w_{1:t} \in V^+$, $p(w_{1:t}) \geq 0$
- $\sum_{w_{1:t} \in V^+} p(w_{1:t}) = 1$.

$p(w_1 \mid w_{1:0}) = p(w_1 \mid \text{EOS}) = p(w_1)$

$|V| = 10^4$: $p(w_{1:5})$ $10^{4 \times 5} = 10^{20}$

$\begin{cases} \text{Statistical language model: N-gram} \\ \text{Neural language models: RNNs (LSTMs, GRUs)} \\ \text{LLMs: Transformer-based.} \end{cases}$

$$\max_{\theta} \frac{1}{N} \sum_{i=1}^{N} \log p_\theta(w^{(i)}) \qquad s_i = w^{(i)} \text{ sentence}$$

$D_{tr} = \{s_1, s_2, \cdots, s_N\}$. i.i.d

$$p_\theta(s_{1:N}) = \prod_{i=1}^{N} p_\theta(s_i) = \prod_{i=1}^{N} \prod_{j=1}^{t_i} p_\theta(w_j^i \mid w_{j-1}^i)$$

$$= \prod_{i=0}^{|V|} \prod_{j=0}^{|V|} p_\theta(v_{j+1} \mid v_i)^{c(v_i, v_{j+1})}$$

$c(v_i, v_{j+1})$: # of bigrams (v_i, v_{j+1}) appeared in

s.t. $\{ v_0 = BOS$

$\theta_{i,j+1} := p_\theta(v_{j+1} \mid v_i)$

$\theta_i = [\theta_{i1}, \theta_{i2}, \cdots, \theta_{i,|V|+1}]^T$. for $i = 0, 1, \cdots, |V|$

For all $i = 0, 1, \cdots, |V|$

$$\sum_{j=0}^{|V|} \theta_{i,j+1} = \sum_{j=0}^{|V|} p_\theta(v_{j+1} \mid v_i) = 1.$$

$$\log p_\theta(s_{1:N}) = \sum_{i=0}^{|V|} \sum_{j=0}^{|V|} c(v_i, v_{j+1}) \log p_\theta(v_{j+1} \mid v_i)$$

$$\Rightarrow \max_{\theta_{0:|V|}} \log p_\theta(s_{1:N} ; \theta_{0:|V|})$$

subject to, $\sum_{j=0}^{|V|} \theta_{i,j+1} = 1, \ \forall i = 0, 1, \cdots, |V|.$

$\Rightarrow$ Lagrange multipliers λ_i ...

$$\theta^*_{i,j+1} = p_\theta(v_{j+1} \mid v_i) = \frac{c(v_i, v_{j+1})}{\sum_{j=0}^{|V|} c(v_i, v_{j+1})}$$

$\Rightarrow$ generalize:

$$p_\theta(x_N \mid x_{1:N-1}) = \frac{c(x_{1:N})}{c(x_{1:N-1})}$$

PPL: From test dataset: $D_{test} = \{s_{1:m}\}$

$$\underset{s_i}{\text{maximize}} \log p_\theta(s_i)$$

$$\prod_{i=1}^{m} p_\theta(s_i) \Leftrightarrow \sum_{i=1}^{m} \log_2 p_\theta(s_i)$$

$\Leftrightarrow$ averaging: $\frac{1}{\sum_{i=1}^{m} |s_i|} \sum_{i=1}^{m} \log_2 p_\theta(s_i) := \ell.$

$$PPL(s_{1:m}) := 2^{-\ell}, \ \ell$$

Note: $p_\theta(s_{1:m})^{-1/T} = \exp\left(-\frac{1}{T} \log p_\theta(s_{1:m})\right)$

$$\left(\prod_{i=1}^{m} p_\theta(s_i)\right)^{\frac{-1}{T}} \qquad T := \sum_{i=1}^{m} |s_i|$$

Connection to entropy and cross-entropy

$$H(p) \triangleq -\sum_{i=1}^{n} p(x_i) \cdot \log p(x_i) \qquad X: x_1, x_2, \cdots, x_n$$

$$H(p,q) \triangleq -\mathbb{E}_p[\log q]$$

$$= -\sum_{x_i \in X} p(x_i) \log q(x_i)$$

$$= -\sum_{x_i \in X} p(x_i)[\log p(x_i) + \log q(x_i) - \log p(x_i)]$$

$$= -\sum_{x_i \in X} p(x_i) \cdot \log p(x_i)$$

$$+ \underbrace{\sum_{x_i \in X} p(x_i)[\log p(x_i) - \log q(x_i)]}_{D_{KL}(p,q) \geq 0}$$

$$\geq H(p).$$

$$H(p) \leq H(p,q) = \lim_{n \to \infty} -\frac{1}{n} \log q(w_{1:n})$$

$$\approx H(s_{1:m}) \triangleq -\frac{1}{|s_{1:m}|} \cdot \log p(s_{1:m})$$

$$PPL(s_{1:m}) := 2^{H(s_{1:m})}.$$

Good-Turing :

N_r : # of types that occur exactly r times

$$r^* = \frac{(r+1)N_{r+1}}{N_r}, \quad P^*_{GT}(w) = \frac{r^*}{N}$$

$$\frac{r^*}{N} = \frac{(r+1)N_{r+1}}{N_r \cdot N}$$

N_r : such words, so that total mass assigned to all words with count r is

$$N_r \cdot \frac{r^*}{N} = N_r \cdot \frac{(r+1)\cdot N_{r+1}}{N_r \cdot N} = \frac{(r+1)\cdot N_{r+1}}{N}$$

$$(*) = \sum_{r \geq 0} \sum_{w: c(w)=r} P^*_{GT}(w) = \sum_{r \geq 0} \frac{(r+1)\cdot N_{r+1}}{N}$$

Let $s = r+1$

Total mass

$$(*) := \sum_{r \geq 0} \frac{(r+1)\cdot N_{r+1}}{N} = \frac{1}{N} \sum_{s \geq 1} s \cdot N_s$$

N_s : is the number of types occuring s times, so

$$\sum_{s \geq 1} s \cdot N_s = N. \quad \Rightarrow (*) = 1.$$

p^* is a distribution.